Software Requirements Specification

for

application for keeping track of watering plants

GreeLendar

Version final

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Table of Contents

Revision History

Name	Date	Reason for Changes	Version
Wiktor Mandra	01.05.2026	Functional and non-functional requirements definition	1.0
Wiktor Mandra	09.05.2026	Use case diagram	2.0
Wiktor Mandra	22.05.2026	Class diagram	3.0
Wiktor Mandra	12.06.2026	Updated class diagram; system features	4.0
Wiktor Mandra	12.06.2026	Diagrams: Sequence, State, Activity	5.0

1. Introduction

1.1. Purpose

This document specifies software requirements for “GreeLendar”, a mobile application designed to help users track the watering, fertillization and genereal care of their plants. This app aims to simplify plant maintenance by providing reminders, logs and care recommendations based on plant species, environment, and user input.

1.2. Document Conventions

1.2.1. Terminology:

Plant Profile: A digital record of a user’s plant including species, age and care history.

Care Log: A timeline of actions (watering, fertillization, etc.) performed on a plant.

1.2.2. Formatting:

Requirements are labeled as REQ-xx (functional) or NF-xx (non-functional).

Priorities: High (H), Medium (M), Low (L).

1.3. Intended Audience and Reading Suggestions

Developers: Focus on technical constriants and interfaces, scetions ….

Project Managers: Review scope, dependencies, and timelines, sections ….

Users: Understand features and user interactions, sections ….

Testers: Verify functional and non-functional requirements, senctions ….

1.4. Product Scope

GreeLendar is a cross-platform app (Android/iOS/Web) that:

Tracks watering, fertilization and repotting schedules.

Provides care reminders based on plants species and evironmental conditions (e.g., humidity, sunlight, temperature).

Logs historical data (e.g., growth photos, soil moisture).

Offers a database of plant care guides (e.g., water frequency, ideal light).

Supports multi-user households (shared plant care).

1.5. References

(1998). *IEEE Recommended Practice for Software Requirements Specifications* (IEEE Std 830-1998) [Review of *IEEE Recommended Practice for Software Requirements Specifications*]. IEEE. https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=720574

Trefle | The plants API. (2024). Trefle.io. https://trefle.io/

Google. (2023). *Material Design*. Material Design. https://m3.material.io/

2. Overall Description

2.1. Product Perspective

The GreeLendar is a standalone app which utilizes third party Trefle API and PostrgreSQL for data backup and between user synchornization.

2.2. Product Features

2.2.1 Main Features

Plant profile: Add or edit plants with species name, photo, care guidelines, log information

Care scheduling: Set cyclical reminders for action such as watering, repoting, etc.

Log actions: Save histiorical action taken to specific plant

Plants Database: Browse list of all plats list provided by Trefle

Take picture: Take picture when adding new plant or to confirm action taken

2.3. Operating Environment

Mobile: iOS15+, Adnroid 10+

Web: Chrome, Firefox, Safari (latest 2 versions)

Backend: Node.js + SQLite

API: Trefle 1.6.0

2.4. Assumptions and Dependencies

Assumptions:

Users have basic plant knowledge: name and watering frequency

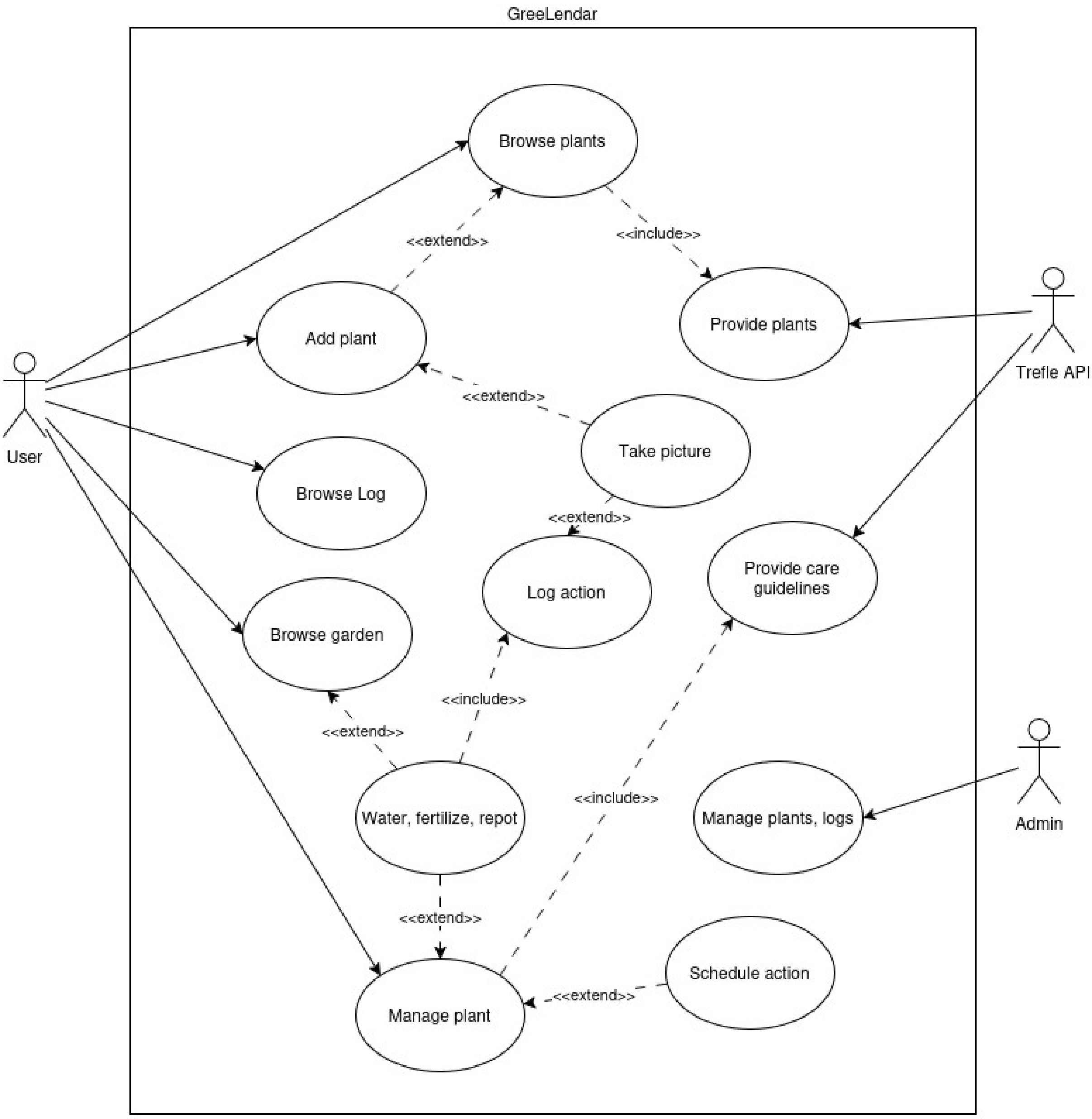
App has access to internet connection at least when adding new plant

Dependencies:

Trefle.io API to retrieve plant list when adding new plant to garden

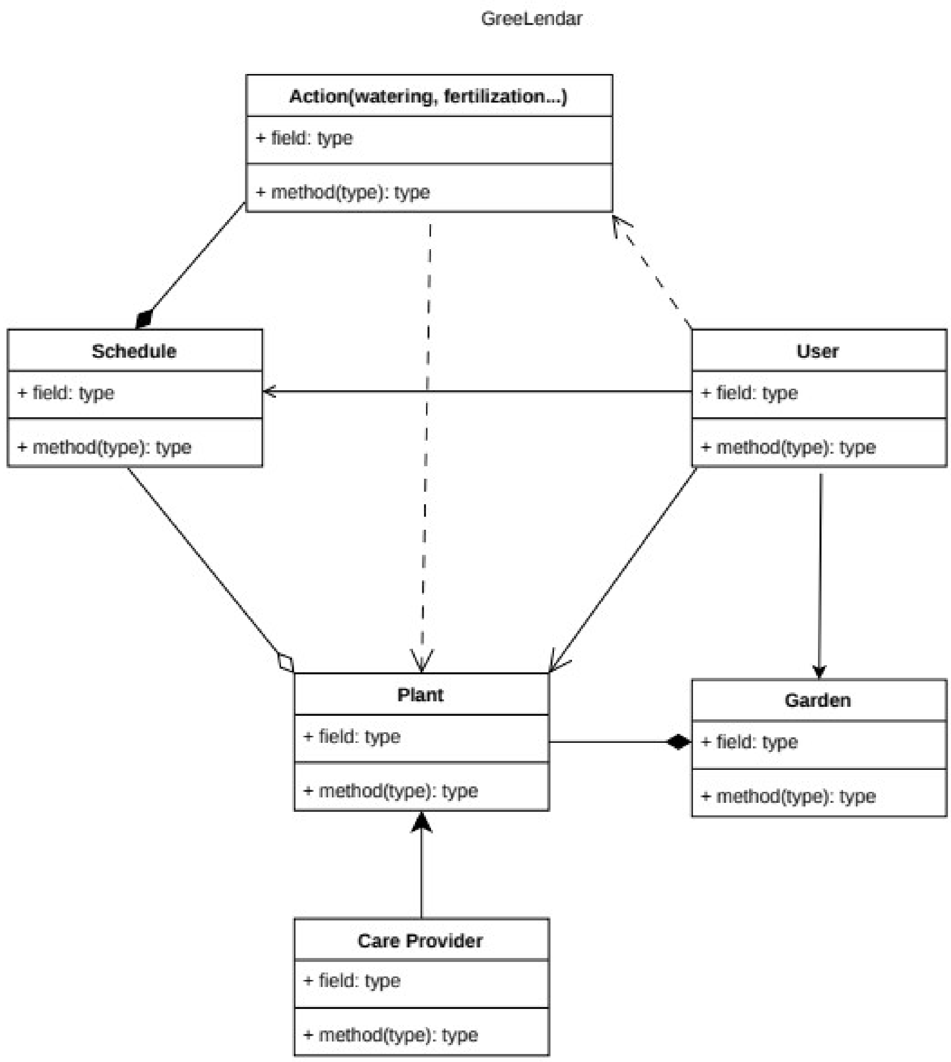
3. Diagrams

3.1. Use Case Diagram



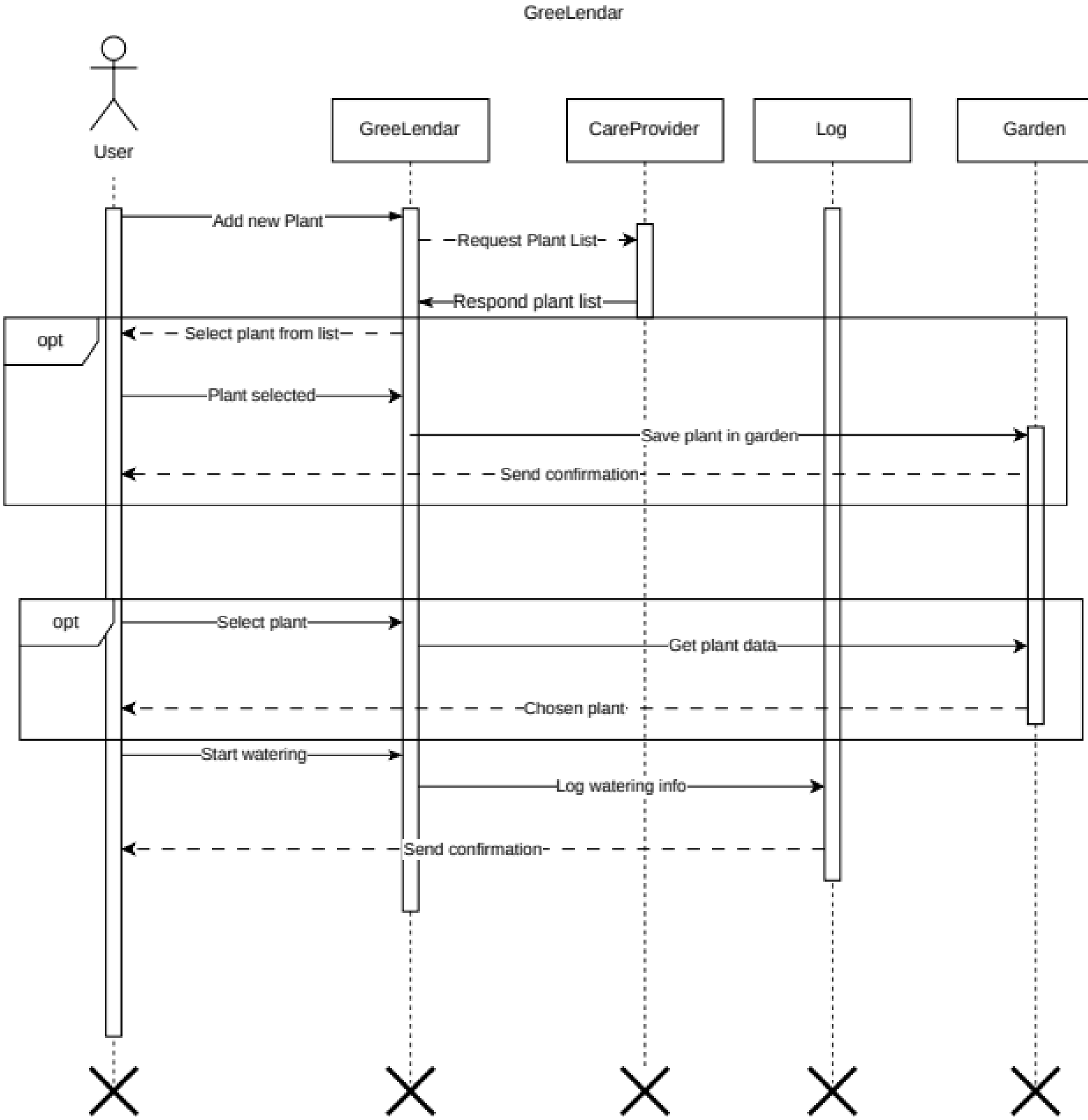
The GreeLendar plant care system is designed to help and guide users in plantcare routines. It enables them to add plants to virtual garden schedule watering etc., check guidelines. The Trefle API porived all data regarding plants and how to care for them. Admin can manage all the actions.

3.2. Class Diagram



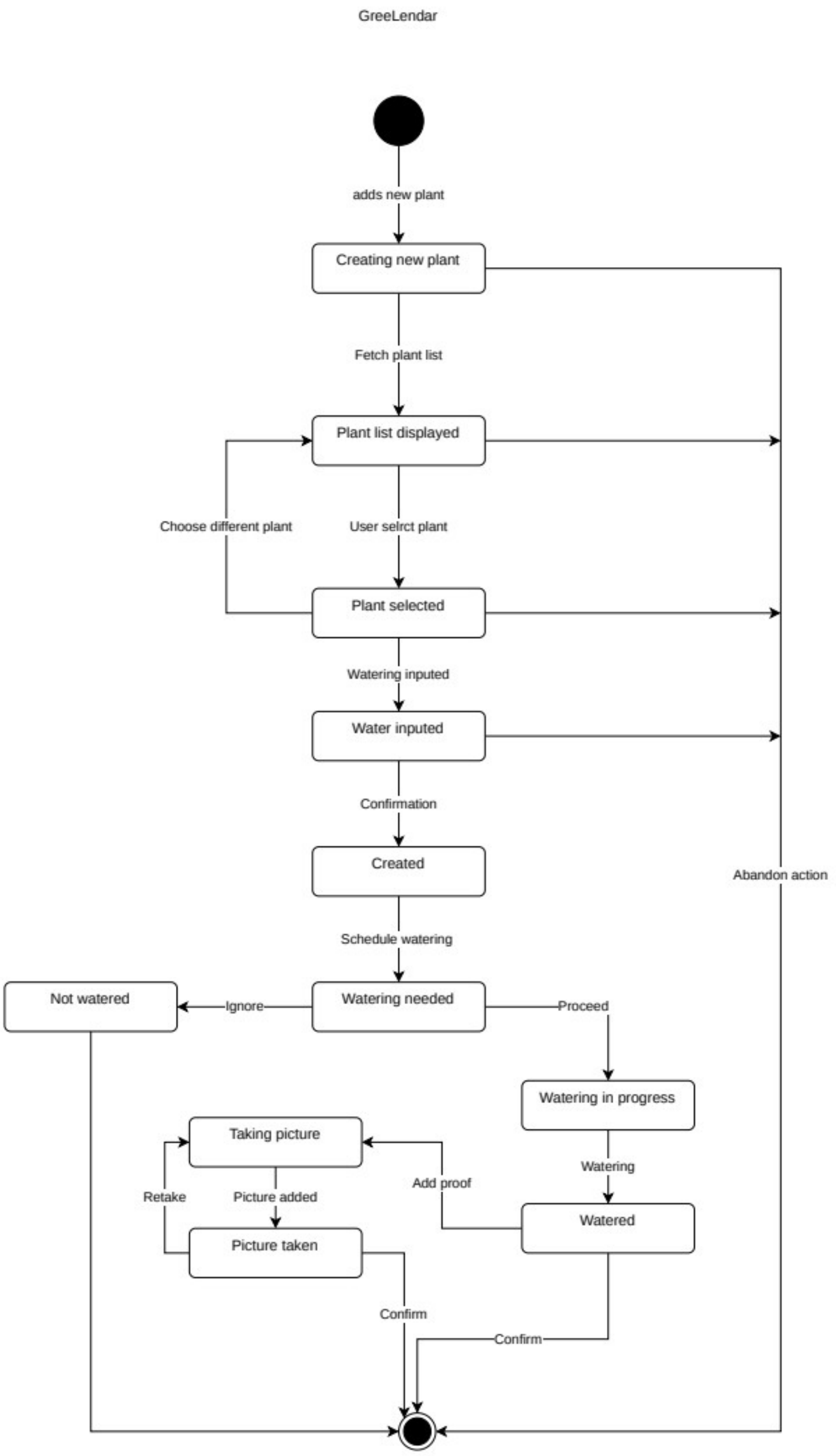
Greenlendar class diagram models a structure plant watering app. User adds Plants, creates Gardens and Schedules. He can take Action such as watering. Plant has its Schedule of watering. Garden contains Plants. Plats are provided by CareProvider. Schedule contains previous Actions.

3.3. Sequence Diagram



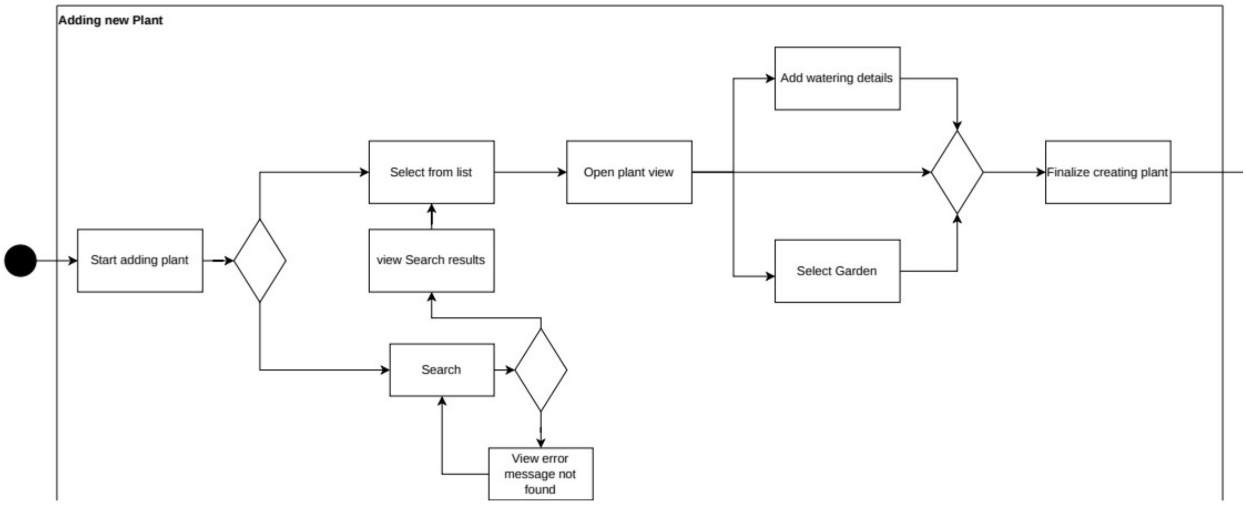
Sequence Diagaram for GreeLendar show flow of action from adding new plant to garden to watering it. User needs to select from plats provided by CareProvider and later choose a plant to be watered from Garden

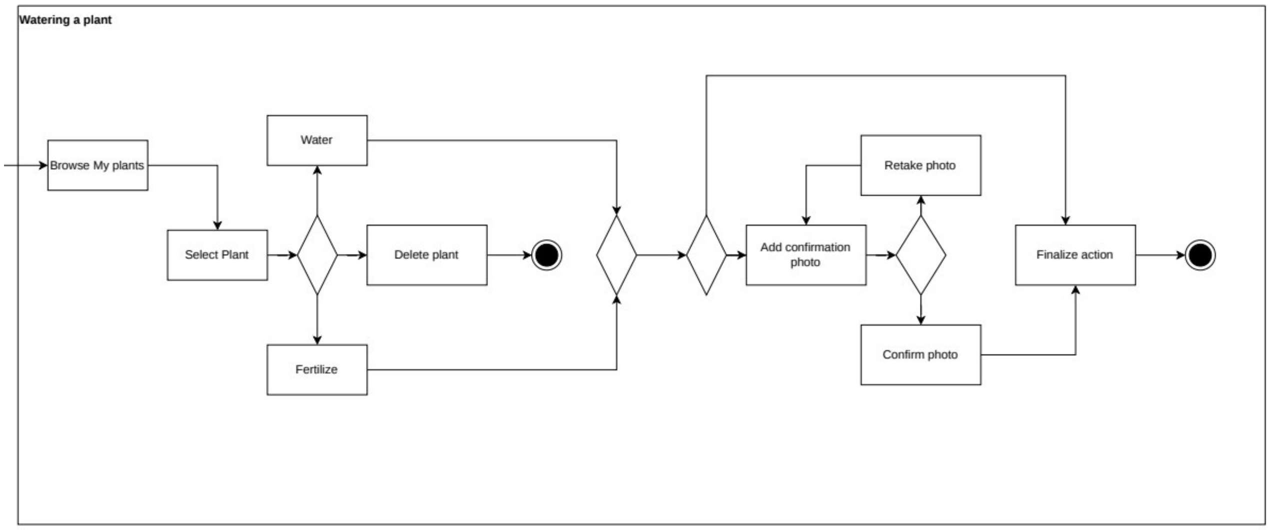
3.4 State Diagram



State Diagram of greeLenar goes through the plant creation process and watering it includes choosing plants from the provided list and take picture to record watering action

3.5 Activity Diagram





GreeLendar Activity diagram show setps to be taken in order to add new plant and then to water it or fertilize

3. System Features

Short overview of features in GreeLendar

3.1. Manage Plant

Add, edit, delete plant. Group them into gardens

3.2. Action Schedule

Schedule when to water, fertailize or sprinkle plants.

3.3. Logging action history

Every action taken upon plant is stored in database and user can view hitorical actions

3.4 Multi-user collaboration

Allows users to share their plants or gardens and assign care to another user

3.5 Notifications

When scheduled action time comes app sends push notificaton to alert user

4. Requirements

4.1. Functional Requirements

REQ-1.1: The app shall allow users to add plant by providing at least species name

REQ-1.2: The app shall support uploading (JPEG/PNG) images for plant profiles

REQ-2.1: The app shall allow users to set custom intervals for each care task

REQ-2.2: The app shall send notification 1 hour before the scheduled task

REQ-3.1: The app shall log date, time, and user for each care action

REQ-3.2: The app shall display timeline of past actions for each plant.

4.2. Nonfunctional Requirements

4.2.1. Performance

NF-1.1: The app shall load dashboard in <2 seconds

NF-1.2: The app shall support up to 100 plants per user without performance issues

4.2.2. Security

NF-2.1: The app shall require authentication for all sensitive acions (e.g., deleting plants)

4.2.3. Usability

NF-3.1 The app shall support dark mode